

Public Perceptions of Algorithmic vs. Human-Drawn Congressional Districts: A Preregistered Survey Protocol

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Evidence status: protocol only. This document does not report a completed human-subjects study. All effect sizes, p -values, respondent characteristics, exclusions, balance checks, and robustness results are synthetic illustrative values. They must not be cited as survey findings.

Abstract

Gerrymandering is widely condemned by the American public, yet political reform remains slow. This paper specifies a proposed survey experiment ($N = 2,400$ respondents) testing perceptions of a partisan-input-excluded geographic benchmark, commission-drawn maps, and enacted maps under different information treatments. No respondents have been recruited, no study has been preregistered, and no empirical result is reported. Numerical estimates in the worked analysis are synthetic placeholders used to test table structure, power, and the planned interpretation protocol. The study must be registered, fielded, archived, and independently reproducible before any claim about public legitimacy, algorithm aversion, partisan moderation, or treatment effects may be made.

Keywords: redistricting, gerrymandering, survey experiment, algorithm aversion, procedural fairness, public opinion

1 Introduction

Americans across partisan lines have long condemned gerrymandering. Surveys consistently find that majorities of Democrats, Republicans, and independents view partisan manipulation of district lines as unfair (Pew Research Center, 2017; Gallup Organization, 2019). Yet the gap between public disapproval and institutional reform is wide and persistent. As of the 2020 redistricting cycle, most states continued to assign line-drawing authority to legislatures with undisguised partisan interests in the outcome. The Supreme Court’s decision in *Rucho v. Common Cause* (2019) removed the federal judiciary as a check on partisan gerrymandering, leaving structural reform to state legislatures, citizen initiatives, and state courts—all of which move slowly.

One structural alternative has received increasing scholarly and journalistic attention: an algorithmic geographic benchmark. The benchmark can exclude partisan data from execution and publish its inputs and parameters, but it is not value-free or structurally immune to manipulation. Profile selection, geographic resolution, data construction, and downstream modifications remain human choices (Della-Libera, 2026).

Despite growing technical validation of algorithmic approaches, a critical question remains unanswered: *do ordinary citizens find algorithmic maps legitimate?* Algorithmic redistricting carries a different kind of risk than legislative or even commission-based redistricting—not the risk of partisan manipulation, but the risk of algorithm aversion, a well-documented tendency for people to discount or distrust algorithmic judgments even when those judgments demonstrably outperform human alternatives (Dietvorst et al., 2015). If the public perceives algorithmic maps as cold, opaque, or undemocratic—as decisions “made by a machine” rather than by accountable humans—then technical superiority may not translate into political

feasibility.

This paper proposes a large-scale survey experiment with $N = 2,400$ respondents recruited from two panel sources. Before recruitment, the research team would register the hypotheses, stimuli, exclusions, stopping rule, analysis code, and interpretation rules. Participants would view algorithmic, commission-drawn, or enacted maps under one of three information conditions.

The protocol tests whether map provenance, plain-language process information, partisanship, and commission comparison affect perceived fairness, process trust, and willingness to accept an unfavorable outcome. The numerical examples in the results section are synthetic and demonstrate only how the preregistered analysis would be reported under one hypothetical outcome.

The paper proceeds as follows. Section 2 reviews the motivating literatures. Section 3 describes the proposed design, sampling, and registration. Section 4 defines planned outcomes. Section 5 provides a synthetic reporting template. Sections 6 and 7 define interpretation boundaries and the evidence required before policy conclusions.

1.1 The Legitimacy Problem in Redistricting Reform

Understanding why reform has been so difficult requires distinguishing between two types of legitimacy that redistricting maps must satisfy. *Outcome legitimacy* concerns whether the resulting districts are judged as fair—whether seat distributions roughly correspond to vote shares, whether district shapes are recognizable as communities, whether minority representation is preserved. *Process legitimacy* concerns whether the procedure that produced the map was itself fair—transparent, neutral, free from self-interested manipulation, and open to public input (Tyler, 2006).

American redistricting reform efforts have focused heavily on process legitimacy: the independent commission model attempts to move mapmaking authority away from self-interested legislatures toward bodies composed of citizens without partisan stakes in the

outcome. Arizona, California, Michigan, and Colorado all adopted commission-based reforms precisely on procedural grounds. The empirical evidence on whether commissions actually produce better outcomes is mixed (Imig and Imig, 2022; Kousser, 2013; Stephanopoulos and Warshaw, 2015): commissions reduce the most egregious partisan bias but do not eliminate it, and they introduce their own procedural tensions around commissioner selection and public input weighting.

Algorithmic redistricting addresses process legitimacy in a structurally different way. Rather than substituting one set of human decision-makers for another, it substitutes a mathematical procedure for human line-drawing. The algorithm cannot be captured by partisan interests because it has no access to partisan data. Its outputs are reproducible—the same inputs always produce the same map—which enables independent verification in a way that commission deliberations cannot. Its decision process is fully documented in open-source code, not in meeting minutes or commissioner testimony.

Whether these procedural properties translate into perceived legitimacy is the empirical question this protocol is designed to test.

2 Related Work

Our study draws on three bodies of literature: public opinion research on redistricting and gerrymandering, procedural fairness theory, and the psychology of algorithm aversion.

2.1 Public Opinion on Redistricting

Public knowledge of and opinion about redistricting has been studied systematically for roughly two decades (Henderson, Haworth and Kim, 2018; Bandyopadhyay and Oak, 2013; Clark and Leiter, 2021). Several consistent findings emerge. First, most Americans have low baseline knowledge of how congressional districts are drawn: surveys typically find fewer than one in three respondents can correctly identify who draws congressional dis-

strict lines in their state (Gimpel and Tam Cho, 2020; Henderson, Haworth and Kim, 2018). Second, when provided with descriptions of gerrymandering—including images of distorted districts—majorities across partisan groups rate the practice as unfair (Elmendorf and Quinn, 2019; Nalder, 2019). Third, public preferences for redistricting process strongly favor independent commissions over legislative control, with commission support in the 60–70% range across multiple polls (Pew Research Center, 2017; Gallup Organization, 2019).

Less studied is public opinion about *algorithmic* redistricting specifically. Nalder (2019) found that subjects shown maps described as “drawn by computer” rated them as more fair than maps described as “drawn by the legislature,” but the study used a small convenience sample and did not vary information about the algorithmic process. Henderson, Haworth and Kim (2018) asked respondents whether they would support computer-drawn districts and found plurality support (approximately 48%), but did not experimentally test whether exposure to actual maps or process descriptions affects these preferences. The proposed study is designed to test this gap at larger scale.

The partisan asymmetry in redistricting perceptions is well-documented. Partisans evaluate maps more favorably when those maps benefit their party (Clark and Leiter, 2021; Bonneau and Cann, 2021). This motivated hypothesis testing party as a moderator—if algorithmic maps are perceived as taking away partisan advantage, the out-party (whichever party benefits less from the current enacted map) should show larger positive responses to the algorithmic alternative. Our design tests this by focusing on enacted maps in states with known partisan gerrymanders favoring each party.

Importantly, research on redistricting reform support suggests that institutional features other than partisan advantage also shape opinion. Imig and Imig (2022) showed that transparency and public accountability in commission processes increase support for commission-drawn maps. Stephanopoulos and Warshaw (2015) found that detailed process descriptions—explaining the criteria commissions use and how they apply them—raised fairness ratings substantially. These findings set the stage for our transparency treatment.

2.2 Procedural Fairness Theory

Procedural fairness theory, developed by Thibaut and Walker (1975) and elaborated by Tyler (2006, 1988), holds that people’s evaluations of outcomes are powerfully shaped by their assessments of the *process* that produced those outcomes. Even unfavorable outcomes may be accepted as legitimate if the procedure was perceived as fair—neutral, transparent, voice-providing, and consistent. Conversely, favorable outcomes may be rejected if the procedure was seen as biased or opaque.

Four procedural characteristics consistently emerge as central to legitimacy assessments (Leventhal, 1980; Lind and Tyler, 1988): *consistency* (the same rules applied equally to all); *bias suppression* (decision-makers lack self-interest in the outcome); *accuracy* (decisions are based on correct information); and *correctability* (errors can be challenged and revised). Algorithmic redistricting has structural advantages on the first two dimensions—algorithms are definitionally consistent, and properly designed algorithms can be made free of partisan self-interest. These properties should translate into procedural legitimacy perceptions when subjects understand the process.

Applied to redistricting, Gerber and Lewis (2018) showed that respondents in court-ordered redistricting states rated the resulting maps as more procedurally fair, despite often rating them as less favorable to their party’s interests. This finding supports the prediction that process neutrality, when perceived, generates legitimacy independent of outcome. Our transparency treatment tests whether communicating process neutrality in plain language achieves the same effect in a controlled experimental setting.

A limitation in applying procedural fairness theory to algorithmic contexts is that algorithms may be perceived as *non-human* in ways that undercut the typical legitimacy pathway. Standard procedural fairness research involves human decision-makers; respondents may extend different standards to algorithmic procedures (Svensson, 2022). The voice component of procedural fairness—the belief that one had an opportunity to be heard—is structurally different when the decision-maker is code rather than a commissioner. We address this

concern in the discussion.

2.3 Algorithm Aversion

A substantial literature documents *algorithm aversion*: the tendency for people to distrust algorithmic judgments and prefer human judgment even when algorithms demonstrably outperform humans (Dietvorst et al., 2015; Dietvorst, Logg and Massey, 2018; Logg, Minson and Moore, 2019). Dietvorst et al. (2015) showed that subjects who watched an algorithm make errors subsequently preferred human forecasters over the algorithm, even when the algorithm’s predictions were objectively more accurate. The effect appears to stem from a belief that algorithms are rigid and cannot adapt to exceptional cases—a concern that human experts, by contrast, can exercise contextual judgment.

Several factors moderate algorithm aversion. Logg, Minson and Moore (2019) showed that algorithm aversion reverses to *algorithm appreciation* when subjects are experts in the relevant domain—that is, experts trust algorithms more than novices do. Dietvorst, Logg and Massey (2018) found that allowing subjects to adjust algorithmic outputs substantially reduced aversion, presumably because it preserved the sense of human agency in the final outcome. Castelo, Bos and Lehmann (2019) found that algorithm aversion is stronger for tasks perceived as requiring uniquely human capacities (e.g., social judgment) than for tasks perceived as objective computation. Redistricting, as a geographic optimization problem with mathematically specifiable criteria, may be perceived as closer to the latter category.

The existing algorithm aversion literature is largely silent on political redistricting. The one domain closest to our context is algorithmic hiring and personnel decisions, where Lee (2018) found that perceived fairness of algorithmic decisions was sharply moderated by transparency: subjects who received explanations of how hiring algorithms worked rated them as substantially more fair than subjects who received only the output. This transparency-fairness link motivates our information condition manipulation.

Our study makes three contributions to the algorithm aversion literature. First, we test

whether aversion appears in a politically salient domain where the relevant comparison is not algorithm-vs.-human expert but algorithm-vs.-politically-motivated legislature. Second, we test whether plain-language process descriptions reduce or eliminate any aversion that appears. Third, we test partisan moderation of aversion—whether political stakes create systematic differences in algorithm evaluations between partisan groups.

2.4 Hypotheses

Drawing on these three literatures, the planned registration will specify the following hypotheses:

H1 (Main effect): Respondents will rate algorithmic maps as fairer than enacted maps.

H2 (Transparency): Respondents will rate algorithmic maps no less fair than commission-drawn maps (null hypothesis of equivalence, tested via one-sided *t*-test). A plain-language description of the algorithmic process will increase perceived fairness for algorithmic and commission maps relative to the enacted baseline.

H3 (Exploratory): We explore whether partisan identification moderates fairness perceptions. Given per-subgroup sample sizes of $n \approx 130\text{--}150$, this analysis is underpowered to detect effects smaller than 0.36 SD at 80% power and should be treated as descriptive.

H4 (Algorithm vs. commission): Algorithmic and commission-drawn maps will not differ significantly in perceived fairness.

H5 (Process trust): Trust in the redistricting process will be higher for algorithmic than enacted maps, with transparency amplifying this effect.

3 Proposed Research Design

3.1 Experimental Design

The proposed study compares three map provenances:

1. a partisan-input-excluded geographic benchmark;
2. a commission-drawn map; and
3. an enacted legislative map.

The information treatment has three levels: no process description, a plain-language process description, and a description plus comparison. The final registration must use a complete factorial design or state and justify which cells are omitted. The earlier draft mixed a seven-cell layout with a “ 2×3 ” label; that design is superseded.

The target sample is $N = 2,400$ respondents. Final cell sizes will be fixed by an a priori power analysis after the primary contrast, clustering unit, multiple-testing family, and expected attrition are specified.

3.2 Stimulus Governance

Before registration, the study will publish:

- the legal provenance and effective date of every enacted or commission map;
- the complete benchmark profile, manifest, and assignment hash;
- image-generation code, image hashes, dimensions, colors, labels, and accessibility checks;
- the state-selection rationale and any partisan-direction balancing rule; and
- cognitive-testing results for every process description.

The benchmark description must not say that the algorithm finds the “most compact” map, guarantees fairness, or cannot be manipulated. It may state only the inputs excluded from execution, the geographic objective, and the reproducibility procedure.

3.3 Recruitment And Ethics

The proposed recruitment plan uses two independent panel sources with a target of 1,200 respondents each. Quotas, weighting, compensation, eligibility, duplicate prevention, bot screening, and panel-specific exclusions must be registered before launch.

No recruitment may begin until the project records the applicable human-subjects ethics determination, informed-consent language, privacy plan, retention period, and de-identification procedure.

3.4 Registration

The OSF or equivalent registration must freeze:

- hypotheses and confirmatory contrasts;
- the primary and secondary outcomes;
- sampling and stopping rules;
- exclusions and missing-data treatment;
- regression, clustering, weighting, and multiplicity procedures;
- subgroup analyses;
- stimuli and source hashes; and
- the interpretation rules for null, heterogeneous, and adverse results.

No registration currently exists. A real URL and timestamp will replace this statement only after the frozen package is public.

3.5 Procedure And Exclusions

The planned survey will collect demographics and baseline attitudes, present the assigned stimulus under a minimum display time, collect map-specific outcomes, administer attention and manipulation checks, and provide a debriefing.

Exclusion thresholds will be fixed before launch. The final report will show recruitment, exclusions, attrition, and analytic sample sizes by condition. There are currently no recruited or excluded respondents.

4 Planned Measures

4.1 Primary Outcome

The proposed primary outcome is a multi-item perceived-fairness scale administered after each map. Candidate items ask whether boundaries appear reasonable, communities appear equally treated, the map appears manipulated, the process inspires trust, and the districts reflect recognizable geography.

Item wording, response options, reverse coding, missing-item rules, scale construction, and standardization will be finalized through cognitive testing and a real pilot. No reliability, factor-analysis, or pilot estimate is currently available.

4.2 Secondary Outcomes

Planned secondary outcomes include:

- perceived partisan advantage;
- process trust;
- willingness to accept an outcome that disadvantages the respondent's party;
- support for using a benchmark within a commission or court process; and

- understanding of the benchmark’s limits.

The last measure is necessary because a treatment that increases trust by causing respondents to believe an algorithm guarantees fairness would be a misleading communication effect rather than evidence of informed legitimacy.

4.3 Moderators

Preregistered moderators may include party identification, in-party status, political knowledge, education, state familiarity, baseline institutional trust, and algorithm attitudes. Subgroup estimates will be labeled exploratory unless the study is powered and corrected for the full moderator family.

4.4 Manipulation And Balance Checks

Manipulation checks will test whether respondents understood the stated map provenance and the benchmark’s actual claim boundary. Balance tables and exclusion summaries will be generated from observed data.

No manipulation success rate, randomization balance result, Cronbach’s alpha, or robustness result is reported in this protocol.

5 Synthetic Reporting Template

This section contains no empirical survey results. The numerical values in the earlier draft—including 0.41 SD, 0.22 SD, 0.18 SD, p -values, support percentages, balance statistics, and robustness checks—were synthetic worked examples and are withdrawn as findings.

A completed study will report:

1. respondent flow from invitation through the analytic sample;
2. condition counts and deviations from the registered stopping rule;

3. primary contrasts with effect sizes, confidence intervals, exact p -values, and multiplicity adjustments;
4. panel-source heterogeneity and preregistered subgroup estimates;
5. measurement reliability and manipulation-check results;
6. null, adverse, and unexpected findings;
7. code and de-identified data sufficient to reproduce every table; and
8. sensitivity to weights, exclusions, clustering, and missing-data rules.

Each respondent is expected to rate two maps, so a target of 2,400 respondents may produce up to 4,800 respondent-map observations. These are different quantities and must be labeled separately in every table.

No coefficient, confidence interval, partisan difference, or policy-support percentage may be inserted into this section before the registered analysis is executed on archived observed data.

6 Interpretation Plan

The study is designed to distinguish several possible outcomes rather than confirm a preferred legitimacy narrative.

- A benchmark advantage over enacted maps could reflect trust in reproducibility, dislike of legislatures, visual differences, treatment wording, or demand effects.
- Equivalence with commission maps would not show that algorithms and commissions are democratically interchangeable.
- A transparency effect would be normatively useful only if respondents also understand that the benchmark does not guarantee fairness, VRA compliance, community preservation, or partisan symmetry.

- Partisan heterogeneity could be larger among political elites or when respondents evaluate their own districts.
- Null or negative results would be informative evidence about adoption barriers and must be reported.

The design must also test adversarial messages that accurately describe parameter discretion, geographic sorting, legal modifications, and known evidence gaps. A treatment that works only when these limitations are hidden would not support a durable legitimacy claim.

Results from two states and online panels will not establish national public acceptance. External replication, state-specific studies, and real-stakes deliberative settings would remain necessary.

7 Conclusion

This paper defines a survey protocol; it does not establish that the public prefers algorithmic maps, that transparency increases perceived fairness, or that support crosses partisan lines.

The protocol can become an empirical contribution only after public registration, ethics review, recruitment, data collection, reproducible analysis, and release of the permitted replication materials. Until those gates pass, C.6 may be cited for its proposed design and interpretation rules, not for any numerical or policy finding.

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